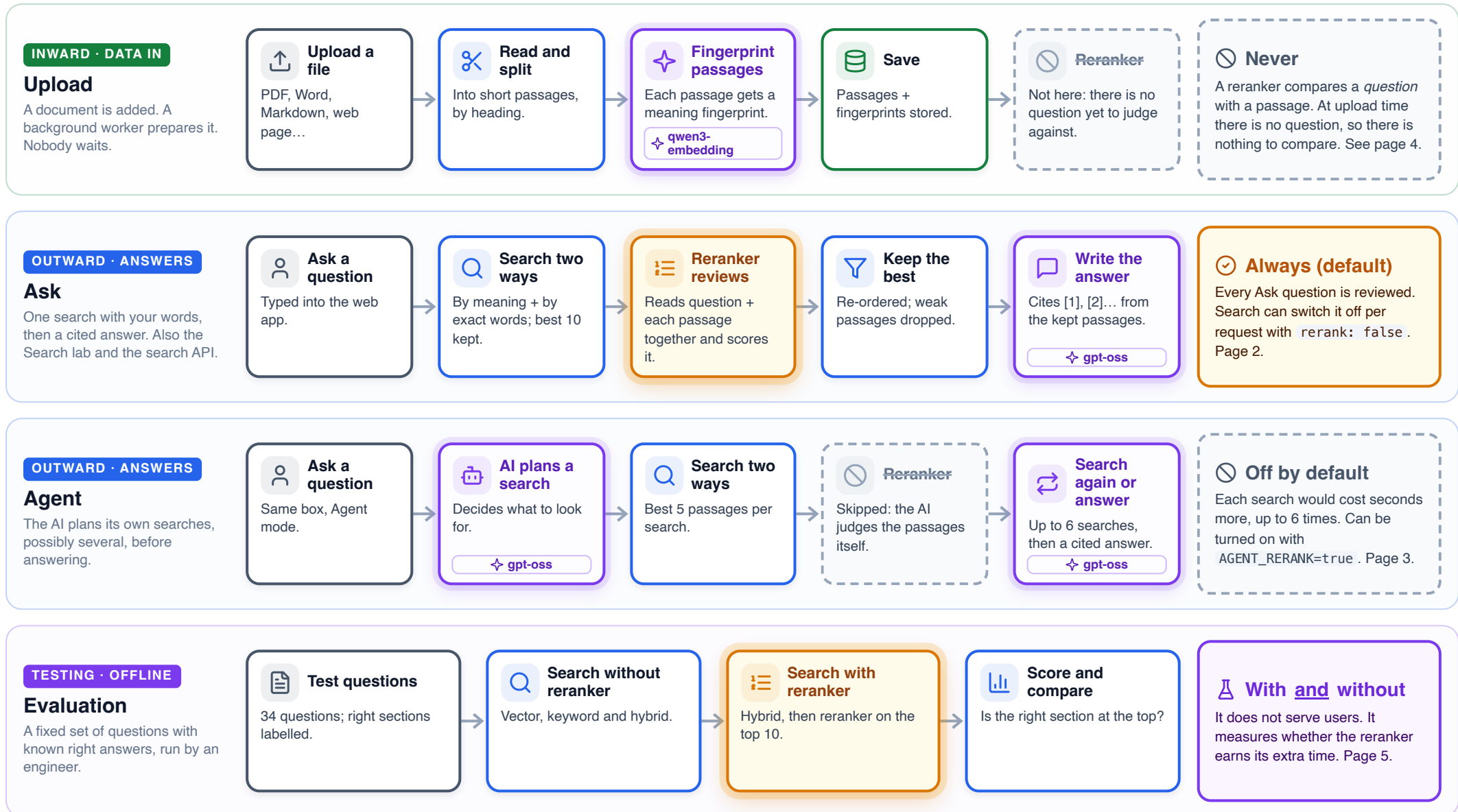


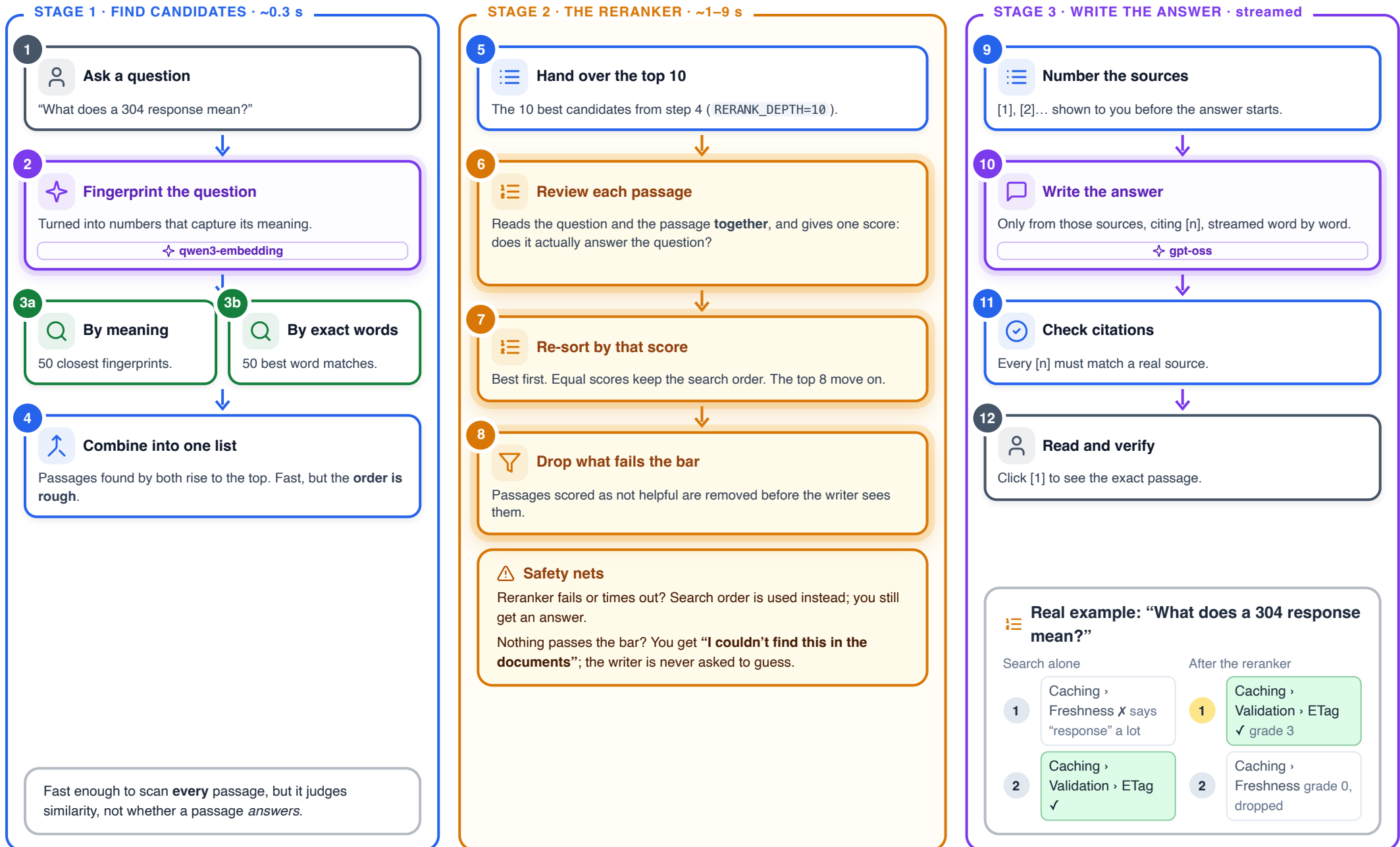
Where the reranker comes in

The **reranker** is a second, careful reviewer. Search finds likely passages quickly; the reranker reads each one next to the question and puts the ones that *really answer it* on top. It needs a question, so it only works on the **outward** side.



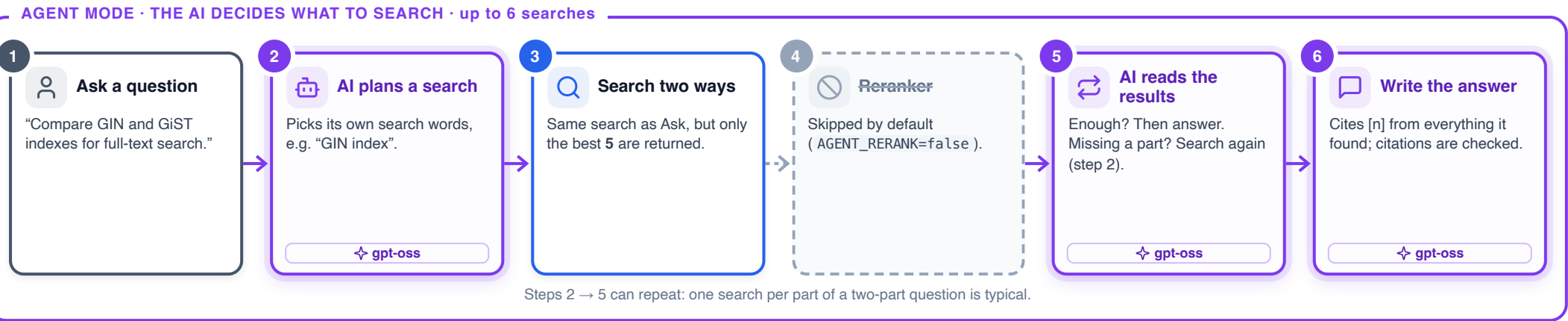
Ask: the reranker sits between search and the answer

On every Ask question, the reranker is **step 6**. It runs once, after the search and before the writer, so the writer only sees passages that were judged to answer the question.



Agent: the reranker is skipped by default

Agent mode uses the same search as Ask, but the chat AI plans the searches and reads the results itself, so the reranker step is left out unless it is switched on.



⌚ Why it is off: time adds up

Ask reranks **once**. The agent may search **up to 6 times**. With the reranker on, each search would wait another 1–9 seconds, so one question could take up to a minute longer.

🔍 Why it is not needed as much

In Agent mode the chat AI **already reads every passage** it gets back and decides whether it is useful. In a way, the agent is its own reviewer.

It also sees fewer passages per search (5, not 10), so there is less noise to sort.

⚙️ It can be switched on

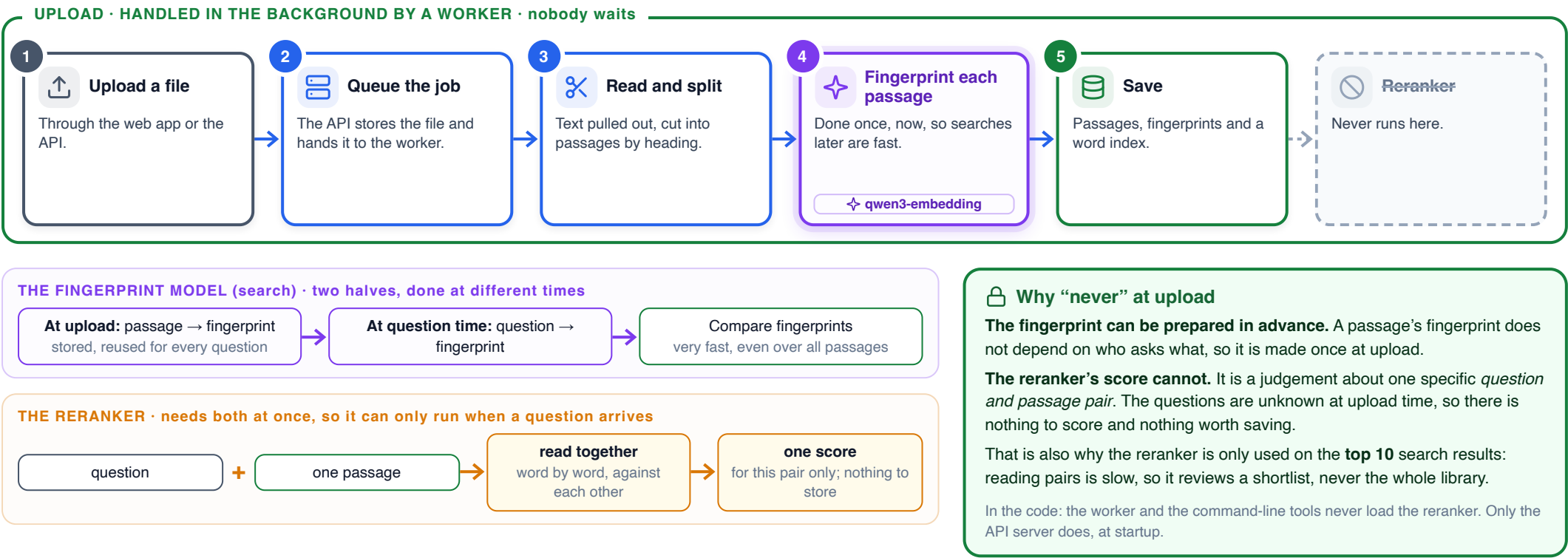
Set `AGENT_RERANK=true`. Every agent search then goes through the reranker, exactly like step 6 on page 2.

Worth trying once the fast cross-encoder reranker is in use (~1–2 s per search instead of up to ~9 s).

	ASK	AGENT
Who decides what to search	The system, using your exact words	The chat AI, in its own words
Searches per question	1	1 to 6
Passages per search	Top 10 reviewed, best 8 kept	Top 5
Reranker	On, every question	Off by default
Who filters weak passages	The reranker's score (drop below the bar)	The chat AI, while reading the results

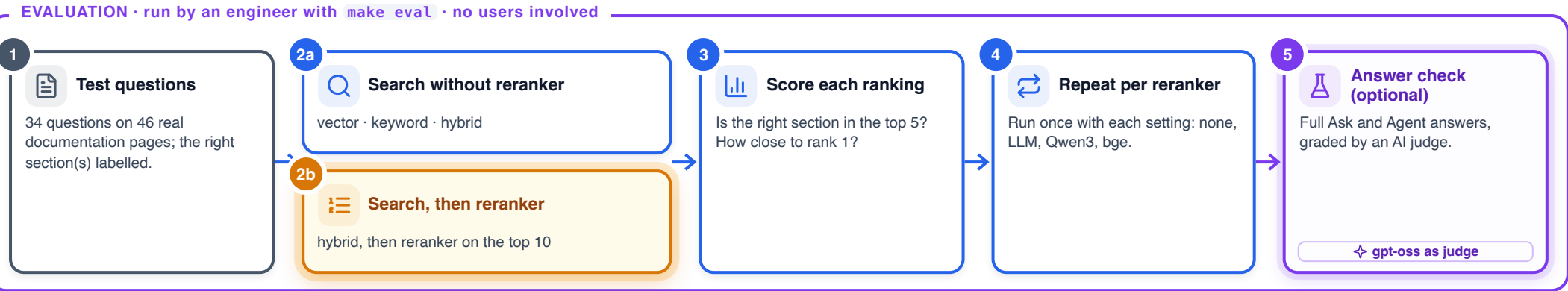
Upload (inward): the reranker is never used

A new document is prepared so it can be searched later. The reranker plays no part: it judges passages *against a question*, and at upload there is none.



Evaluation: measuring whether the reranker helps

The evaluation is a test bench, not a user feature. It runs the same questions through search **with and without** the reranker, so we can see what the extra seconds buy.



Benchmark result: 31 answerable questions

SETUP	FOUND IN TOP 5	RANK-1 SCORE (MRR)	TIME
Hybrid search, no reranker	0.77	0.68	0.3 s
+ LLM reranker (gpt-oss)	0.88	0.91	8.5 s
+ Qwen3-Reranker-0.6B (cross-encoder)	0.88	0.87	2.2 s
+ bge-reranker-v2-m3 (cross-encoder)	0.83	0.84	1.2 s

Both scores run from 0 to 1; higher is better. MRR 1.0 = the right section is always first; 0.5 = typically second. Differences under ~0.04 are within noise for 31 questions.

What it showed

- **Every reranker helps a lot:** the right section reaches rank 1 far more often (MRR 0.68 → 0.84–0.91).
- **Qwen3 ≈ the LLM reranker** in quality, about **4x faster**.
- A reranker can only re-order what search found: if the right passage is not in the top 10, no reranker can fix it.

What the evaluation does not do

It never changes the app or the documents. It reads the same search code real users get, runs it with and without the reranker, and writes a report under `evals/results/`.

For engineers: which reviewer, settings and code

One setting chooses the reviewer. The rest of the app does not care which one: search calls whatever reranker it is given.

☰ Two possible reviewers, one switch

RERANKER=	WHO REVIEWS	SCORE	BAR FOR /ASK	SPEED
llm code default	The chat model gpt-oss (also the writer), grading in batches of 10	0–3	grade ≥ 1	~3–9 s
cross_encoder our .env	Qwen3-Reranker-0.6B , a small dedicated model run inside the API via sentence-transformers	0–1	score ≥ 0.02	~1–2 s
none	Nobody: search order is used as is	—	no filter	0 s

They score on different scales, so each has its own bar. Speeds are for 10 passages on an M1 Max.

⚙️ Settings

SETTING	DEFAULT	EFFECT
RERANK_DEPTH	10	How many search results get reviewed
AGENT_RERANK	false	Also review every agent search
ANSWER_MIN_RERANK_GRADE	1	/ask bar for the LLM reviewer
ANSWER_MIN_CROSS_ENCODER_SCORE	0.02	/ask bar for the cross-encoder
rerank	on	Per-request switch on /search and /ask

<> Where the code lives

FILE	WHAT IT DOES
app/application/ports/reranking.py	The Reranker interface: question + candidates → one score each
app/infrastructure/reranking/llm_reranker.py	The gpt-oss reviewer (grades 0–3)
app/infrastructure/reranking/cross_encoder_reranker.py	The sentence-transformers reviewer (0–1)
app/application/use_cases/retrieval/__init__.py	The only caller: SearchCollection._rerank ; falls back to search order on failure
app/application/use_cases/answering/context.py	Ask only: drops passages below the bar
app/application/use_cases/agent/tools.py	Agent search tool; passes rerank (off by default)
app/bootstrap/container.py	Picks the reviewer and the /ask bar; warms the model up (API only)
app/core/config.py	All reranker settings and defaults
evals/runner.py	Runs search with and without the reranker

More detail: docs/RERANKING.md and docs/EVALUATION.md .